North Deering — Fisher's Combined Probability Test

Fisher's method combines many individual p-values into one overall probability, answering:
'what are the odds of seeing this whole collection of results together by chance?'

Formula: X2 = -2 x sum(ln(p)) ; degrees of freedom = 2 x (number of tests) ;
combined p = upper tail of the chi-square distribution at X2.

Each tab lists every individual p-value that goes into the combination (North Deering's
corrected two-sided Poisson p vs the given baseline), so the calculation is fully traceable.
p-values are the CORRECTED values (two-sided Poisson), not the original file's POISSON-FALSE error.

Caveat: Fisher assumes the tests are independent. These share North Deering's population and
some sites overlap, so they are not fully independent — the combined p is a strong indicator,
best confirmed by a permutation test for formal publication.

Young bands (site-specific)

North Deering vs Portland excl. ND

Age band	Cancer site	ND cases	Expected	p-value	ln(p)
0-19	Acute lymphocytic leukemia	1	1.06	1.0000	0.0000
0-19	Brain	2	0.42	0.1358	-1.9962
0-19	Hodgkin lymphoma	2	0.42	0.1358	-1.9962
20-44	Lung	2	0.82	0.4004	-0.9154
20-44	Colon	2	0.96	0.5008	-0.6916
20-44	Breast (female)	8	5.09	0.2870	-1.2484
20-44	Prostate (male)	1	0.38	0.6348	-0.4545
20-44	Melanoma	9	2.47	**0.0021**	-6.1529
20-44	Thyroid	6	1.51	**0.0093**	-4.6816
20-44	Kidney	3	0.27	**0.0056**	-5.1768

Number of tests (k)	10
X2 = -2 x sum(ln p)	46.627
Degrees of freedom = 2k	20
COMBINED p-value	**0.000661**

North Deering vs Cumberland County

Age band	Cancer site	ND cases	Expected	p-value	ln(p)
0-19	Acute lymphocytic leukemia	1	0.38	0.6272	-0.4664
0-19	Brain	2	0.48	0.1697	-1.7738
0-19	Myeloid & monocytic leukemia	2	0.14	**0.0181**	-4.0098
0-19	Hodgkin lymphoma	2	0.27	0.0612	-2.7934
20-44	Lung	2	1.68	1.0000	0.0000
20-44	Colon	2	1.27	0.7262	-0.3199
20-44	Breast (female)	8	8.25	1.0000	0.0000
20-44	Prostate (male)	1	0.30	0.5218	-0.6506
20-44	Melanoma	9	3.71	**0.0279**	-3.5783
20-44	Thyroid	6	3.02	0.1728	-1.7555
20-44	Kidney	3	1.10	0.1991	-1.6139

Number of tests (k)	11
X2 = -2 x sum(ln p)	33.923
Degrees of freedom = 2k	22
COMBINED p-value	**0.050016**

North Deering vs Maine

Age band	Cancer site	ND cases	Expected	p-value	ln(p)
0-19	Acute lymphocytic leukemia	1	0.35	0.5906	-0.5266
0-19	Brain	2	0.40	0.1231	-2.0947
0-19	Myeloid & monocytic leukemia	2	0.12	**0.0133**	-4.3201

0-19	Hodgkin lymphoma	2	0.19	**0.0318**	-3.4471
20-44	Lung	2	2.10	1.0000	0.0000
20-44	Colon	2	1.30	0.7464	-0.2926
20-44	Breast (female)	8	8.80	0.9646	-0.0361
20-44	Prostate (male)	1	0.30	0.5184	-0.6571
20-44	Melanoma	9	4.26	0.0604	-2.8060
20-44	Thyroid	6	2.96	0.1591	-1.8385
20-44	Kidney	3	0.93	0.1351	-2.0018

Number of tests (k)	11
X2 = -2 x sum(ln p)	36.041
Degrees of freedom = 2k	22
COMBINED p-value	**0.030060**

Young bands (with aggregates)

North Deering vs Portland excl. ND

Age band	Cancer site	ND cases	Expected	p-value	ln(p)
0-19	All cancers	10	4.02	**0.0169**	-4.0833
0-19	Acute lymphocytic leukemia	1	1.06	1.0000	0.0000
0-19	Brain	2	0.42	0.1358	-1.9962
0-19	Hodgkin lymphoma	2	0.42	0.1358	-1.9962
20-44	All cancers	53	27.22	**0.0000**	-11.0631
20-44	Lung	2	0.82	0.4004	-0.9154
20-44	Colon	2	0.96	0.5008	-0.6916
20-44	Breast (female)	8	5.09	0.2870	-1.2484
20-44	Prostate (male)	1	0.38	0.6348	-0.4545
20-44	Melanoma	9	2.47	**0.0021**	-6.1529
20-44	Thyroid	6	1.51	**0.0093**	-4.6816
20-44	Kidney	3	0.27	**0.0056**	-5.1768

Number of tests (k)	12
X2 = -2 x sum(ln p)	76.92
Degrees of freedom = 2k	24
COMBINED p-value	**0.000000**

North Deering vs Cumberland County

Age band	Cancer site	ND cases	Expected	p-value	ln(p)
0-19	All cancers	10	2.40	**0.0004**	-7.8195
0-19	Acute lymphocytic leukemia	1	0.38	0.6272	-0.4664
0-19	Brain	2	0.48	0.1697	-1.7738
0-19	Myeloid & monocytic leukemia	2	0.14	**0.0181**	-4.0098
0-19	Hodgkin lymphoma	2	0.27	0.0612	-2.7934
20-44	All cancers	53	37.22	**0.0172**	-4.0632
20-44	Lung	2	1.68	1.0000	0.0000
20-44	Colon	2	1.27	0.7262	-0.3199
20-44	Breast (female)	8	8.25	1.0000	0.0000
20-44	Prostate (male)	1	0.30	0.5218	-0.6506
20-44	Melanoma	9	3.71	**0.0279**	-3.5783
20-44	Thyroid	6	3.02	0.1728	-1.7555
20-44	Kidney	3	1.10	0.1991	-1.6139

Number of tests (k)	13
X2 = -2 x sum(ln p)	57.688
Degrees of freedom = 2k	26
COMBINED p-value	**0.000340**

North Deering vs Maine

Age band	Cancer site	ND cases	Expected	p-value	ln(p)
0-19	All cancers	10	2.10	**0.0001**	-8.8843
0-19	Acute lymphocytic leukemia	1	0.35	0.5906	-0.5266
0-19	Brain	2	0.40	0.1231	-2.0947
0-19	Myeloid & monocytic leukemia	2	0.12	**0.0133**	-4.3201
0-19	Hodgkin lymphoma	2	0.19	**0.0318**	-3.4471
20-44	All cancers	53	39.22	**0.0412**	-3.1884
20-44	Lung	2	2.10	1.0000	0.0000
20-44	Colon	2	1.30	0.7464	-0.2926
20-44	Breast (female)	8	8.80	0.9646	-0.0361
20-44	Prostate (male)	1	0.30	0.5184	-0.6571
20-44	Melanoma	9	4.26	0.0604	-2.8060
20-44	Thyroid	6	2.96	0.1591	-1.8385
20-44	Kidney	3	0.93	0.1351	-2.0018

Number of tests (k)	13
X2 = -2 x sum(ln p)	60.187
Degrees of freedom = 2k	26
COMBINED p-value	**0.000158**

All ages (site-specific)

North Deering vs Portland excl. ND

Age band	Cancer site	ND cases	Expected	p-value	ln(p)
0-19	Acute lymphocytic leukemia	1	1.06	1.0000	0.0000
0-19	Brain	2	0.42	0.1358	-1.9962
0-19	Hodgkin lymphoma	2	0.42	0.1358	-1.9962
20-44	Lung	2	0.82	0.4004	-0.9154
20-44	Colon	2	0.96	0.5008	-0.6916
20-44	Breast (female)	8	5.09	0.2870	-1.2484
20-44	Prostate (male)	1	0.38	0.6348	-0.4545
20-44	Melanoma	9	2.47	**0.0021**	-6.1529
20-44	Thyroid	6	1.51	**0.0093**	-4.6816
20-44	Kidney	3	0.27	**0.0056**	-5.1768
45-64	Lung	25	25.07	1.0000	0.0000
45-64	Colon	6	6.95	0.9155	-0.0883
45-64	Breast (female)	40	29.50	0.0754	-2.5848
45-64	Prostate (male)	23	18.38	0.3342	-1.0961
65-84	Lung	61	74.57	0.1232	-2.0943
65-84	Colon	32	34.00	0.8180	-0.2009
65-84	Breast (female)	46	37.36	0.1889	-1.6664
65-84	Prostate (male)	41	38.60	0.7409	-0.3000
85+	Lung	15	8.91	0.0771	-2.5627
85+	Colon	10	15.83	0.1665	-1.7928
85+	Breast (female)	12	11.23	0.8953	-0.1106
85+	Prostate (male)	5	1.93	0.0924	-2.3817

Number of tests (k)	22
X2 = -2 x sum(ln p)	76.384
Degrees of freedom = 2k	44
COMBINED p-value	**0.001772**

North Deering vs Cumberland County

Age band	Cancer site	ND cases	Expected	p-value	ln(p)
0-19	Acute lymphocytic leukemia	1	0.38	0.6272	-0.4664
0-19	Brain	2	0.48	0.1697	-1.7738
0-19	Myeloid & monocytic leukemia	2	0.14	**0.0181**	-4.0098
0-19	Hodgkin lymphoma	2	0.27	0.0612	-2.7934
20-44	Lung	2	1.68	1.0000	0.0000
20-44	Colon	2	1.27	0.7262	-0.3199
20-44	Breast (female)	8	8.25	1.0000	0.0000
20-44	Prostate (male)	1	0.30	0.5218	-0.6506
20-44	Melanoma	9	3.71	**0.0279**	-3.5783
20-44	Thyroid	6	3.02	0.1728	-1.7555
20-44	Kidney	3	1.10	0.1991	-1.6139

45-64	Lung	25	20.34	0.3532	-1.0406
45-64	Colon	6	7.82	0.6709	-0.3991
45-64	Breast (female)	40	32.89	0.2520	-1.3782
45-64	Prostate (male)	23	22.77	1.0000	0.0000
65-84	Lung	61	67.92	0.4407	-0.8194
65-84	Colon	32	33.70	0.8583	-0.1528
65-84	Breast (female)	46	45.76	1.0000	0.0000
65-84	Prostate (male)	41	46.85	0.4398	-0.8215
85+	Lung	15	11.11	0.3083	-1.1767
85+	Colon	10	14.79	0.2578	-1.3556
85+	Breast (female)	12	10.85	0.8064	-0.2152
85+	Prostate (male)	5	3.21	0.4427	-0.8149

Number of tests (k)	23
X2 = -2 x sum(ln p)	50.271
Degrees of freedom = 2k	46
COMBINED p-value	**0.308093**

North Deering vs Maine

Age band	Cancer site	ND cases	Expected	p-value	ln(p)
0-19	Acute lymphocytic leukemia	1	0.35	0.5906	-0.5266
0-19	Brain	2	0.40	0.1231	-2.0947
0-19	Myeloid & monocytic leukemia	2	0.12	**0.0133**	-4.3201
0-19	Hodgkin lymphoma	2	0.19	**0.0318**	-3.4471
20-44	Lung	2	2.10	1.0000	0.0000
20-44	Colon	2	1.30	0.7464	-0.2926
20-44	Breast (female)	8	8.80	0.9646	-0.0361
20-44	Prostate (male)	1	0.30	0.5184	-0.6571
20-44	Melanoma	9	4.26	0.0604	-2.8060
20-44	Thyroid	6	2.96	0.1591	-1.8385
20-44	Kidney	3	0.93	0.1351	-2.0018
45-64	Lung	25	24.40	0.9568	-0.0441
45-64	Colon	6	9.60	0.3149	-1.1555
45-64	Breast (female)	40	30.80	0.1260	-2.0718
45-64	Prostate (male)	23	21.10	0.7357	-0.3070
65-84	Lung	61	70.40	0.2873	-1.2471
65-84	Colon	32	36.80	0.4869	-0.7197
65-84	Breast (female)	46	45.40	0.9684	-0.0321
65-84	Prostate (male)	41	57.00	**0.0327**	-3.4192
85+	Lung	15	10.70	0.2501	-1.3860
85+	Colon	10	13.40	0.4378	-0.8259
85+	Breast (female)	12	11.40	0.9367	-0.0653
85+	Prostate (male)	5	3.90	0.7033	-0.3520

Number of tests (k)	23

X2 = -2 x sum(ln p)	59.292
Degrees of freedom = 2k	46
COMBINED p-value	**0.090256**